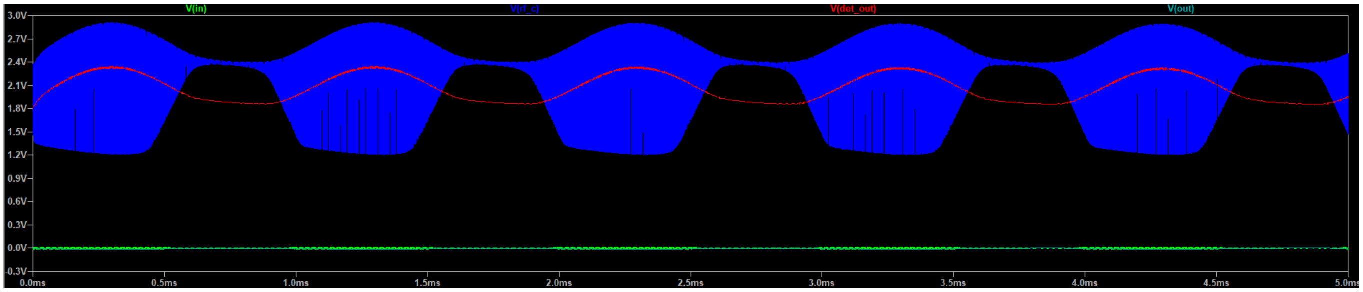
Real_LTSpice_Netlist_3

Prompt



here is what i got from your netlist, the output doesnt seem to appear compared to the RF stage.

Answer

LTSpice AM Radio: Troubleshooting Simulation Output

Based on the screenshot you provided, I can see exactly why your `V(out)` (cyan trace) is a flat line, and I also spot an issue with your RF stage `V(rf_c)` (blue trace). Both are classic analog design challenges.

Here is a breakdown of what is happening and the exact changes you need to make to your netlist.

Problem 1: The Flat Output (Severe Impedance Mismatch)

In your netlist, the final component is `R_load OUT 0 32`, representing 32-ohm headphones.

Your audio amplifier transistor (Q2) uses a 4.7 kΩ collector resistor (`R_af_c`). Because of this, the amplifier has a relatively high "output impedance." When you connect a tiny 32 Ω load directly to it, you create a massive AC voltage divider. The 32 Ω load effectively short-circuits the voltage swing, squashing it down to nearly zero. It is like trying to drive a heavy truck with a bicycle engine.

The Fix: To verify the circuit works mathematically in LTSpice, change the load to represent a high-impedance input (like an oscilloscope probe or the input to a modern audio chip).

- Change: `R_load OUT 0 32`
- To: `R_load OUT 0 10k`

(Note for real-world building: To physically drive 32-ohm headphones from this circuit, you would typically add an LM386 audio amplifier chip as the final stage, which is designed to handle low-impedance speakers).

Problem 2: The Squashed Blue Wave (RF Amplifier Clipping)

Look closely at your blue trace `V(rf_c)`. Notice how the bottom half of the wave is cut off perfectly flat at roughly 1.2V?

Your RF amplifier has a massive amount of gain. The 10mV AM signal we generated is actually too strong. The amplifier is multiplying it so much that the transistor is running out of headroom and slamming into its operational limits (clipping).

The Fix:
Reduce the input signal from 10 millivolts to 1 millivolt to simulate a more realistic, weak antenna signal.

- Change your `B1` line to end with `*1m` instead of `*10m`:
  `B1 IN 0 V=(1+0.8*sin(2*pi*1k*time))*sin(2*pi*693k*time)*1m`

Problem 3: Low Audio Gain (Optional but Recommended)

To get a much stronger voltage swing out of the final audio stage, the audio transistor's emitter needs an AC bypass capacitor (just like the RF stage has).

The Fix:
Add a single capacitor to bypass the audio emitter resistor.

- Add this line right under your `R_af_e AF_E2 0 1k` line:
  `C_af_e AF_E2 0 10u`

The Corrected Netlist

Here is the fully updated netlist with those three fixes applied. You can replace your current file with this code:

```
* AM Radio Receiver Simulation (693 kHz BBC 5 Live) - FIXED
* --- Power Supply ---
Vcc VCC 0 9V

* --- RF Antenna / Signal Source ---
* B1 amplitude reduced to 1mV to prevent RF clipping
B1 IN 0 V=(1+0.8*sin(2*pi*1k*time))*sin(2*pi*693k*time)*1m
```

```
R_ant IN RF_IN 50

* --- LC Tuning Tank ---
L1 RF_IN 0 220u
C_tune RF_IN 0 240p

* --- RF Amplifier Stage ---
C_rf_in RF_IN RF_B 10n
R_rf_b RF_C RF_B 100k
R_rf_c VCC RF_C 4.7k
R_rf_e RF_E 0 1k
C_rf_e RF_E 0 100n
Q1 RF_C RF_B RF_E 2N3904

* --- AM Detector Stage ---
D1 RF_C DET_OUT 1N4148
R_det DET_OUT 0 10k
C_det DET_OUT 0 10n

* --- Audio Amplifier Stage ---
C1 DET_OUT AF_B 10u
R_af_b AF_C AF_B 10k
R_af_c VCC AF_C 4.7k
R_af_e AF_E2 0 1k
C_af_e AF_E2 0 10u      ; Added to boost audio AC gain
Q2 AF_C AF_B AF_E2 2N3904
C2 AF_C OUT 10u
R_load OUT 0 10k        ; Changed to 10k to prevent impedance mismatch

* --- Transistor & Diode Models ---
.model 2N3904 NPN(IS=1E-14 VAF=100 BF=300 IKF=0.4 XTB=1.5 BR=4 CJC=4p CJE=8p
RB=20 RC=0.1 RE=0.1 TR=250p TF=350p ITF=1 VTF=2 XTF=3 VCEO=40)
.model 1N4148 D(Is=2.52n Rs=.568 N=1.752 Cjo=4p M=.4 tt=20n Iave=200m Vpk=75)

* --- Simulation Setup ---
.tran 0 5m 0 100n
.end
```